「Intergrating factor.

Consider an equation $y' + g(x)y = f(x)$.

Let $h(x)$ be a unknown. observe that

$$h(x)\cdot y' + h(x)g(x)y = h(x)f(x).$$

Suppose that $(h(x)y)' = h(x)\cdot y' + h(x)g(x)y$.

Then, since $(h(x)y)' = h(x)y' + h'(x)y$, so $(h'(x) - h(x)g(x))y = 0$.

There fore, $h'(x) = h(x)g(x)$, so

$$\frac{h'(x)}{h(x)} = g(x) \Rightarrow \log|h(x)| = \int g(x) + C \Rightarrow |h(x)| = C_0 \cdot e^{\int g(x)}.$$

「Example.

「Solve $y' + 2y = e^x$.

Proof. Multiplying to the both side $e^{\int 2} = e^{2x}$, then

$$e^{2x}y' + e^{2x}\cdot 2y = e^{3x}$$

Since $e^{2x}y' + e^{2x}\cdot 2y = (e^{2x}y)'$, so

$$e^{2x}y = \int e^{3x}\,dx = \frac{1}{3}e^{3x} + C.$$

$$\Rightarrow y = \frac{1}{3}e^{x} + C\cdot e^{-2x}.$$

「Solve $y' + \frac{1}{x}y = x$, $x > 0$.

Multiplying $e^{\int \frac{1}{x}} = e^{\log x} = x$ to the both side, then

$$xy' + y = x^2 \Rightarrow (xy)' = x^2 \Rightarrow xy = \frac{1}{3}x^3 + C \Rightarrow y = \frac{1}{3}x^2 + \frac{C}{x}.$$